

Applied Text Mining in Python

Generative models and LDA

Generative Models for Text

$\text{Pr}(\text{text} \mid \text{model})$

the 0.1
is 0.07
harry 0.05
potter 0.04
movie 0.04
plot 0.02
time 0.01
rowling 0.01

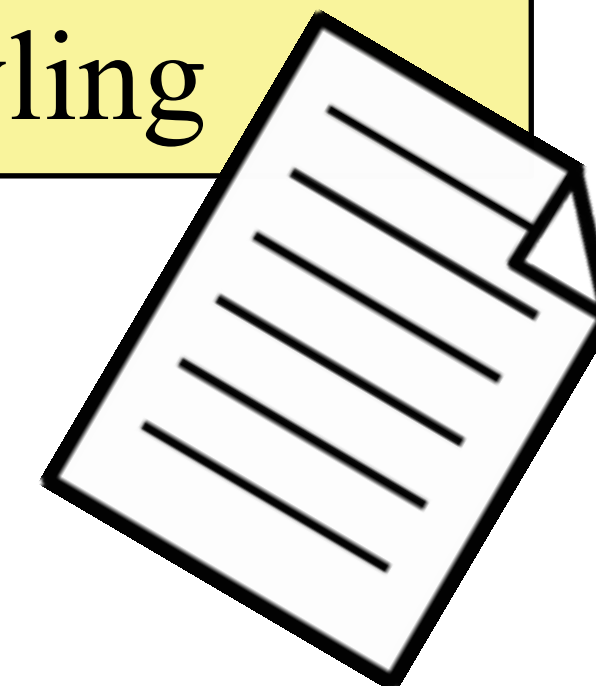
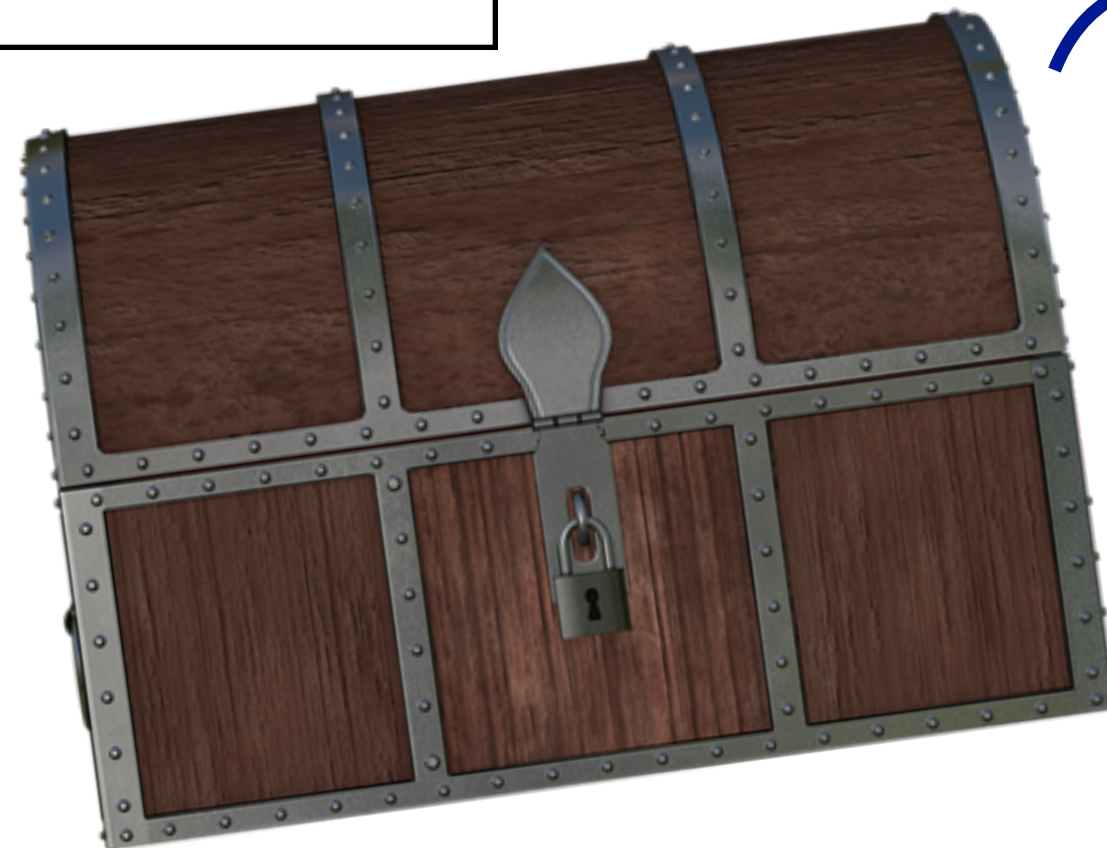
Inference, Estimation



the
harry
potter
movie
is
harry
is

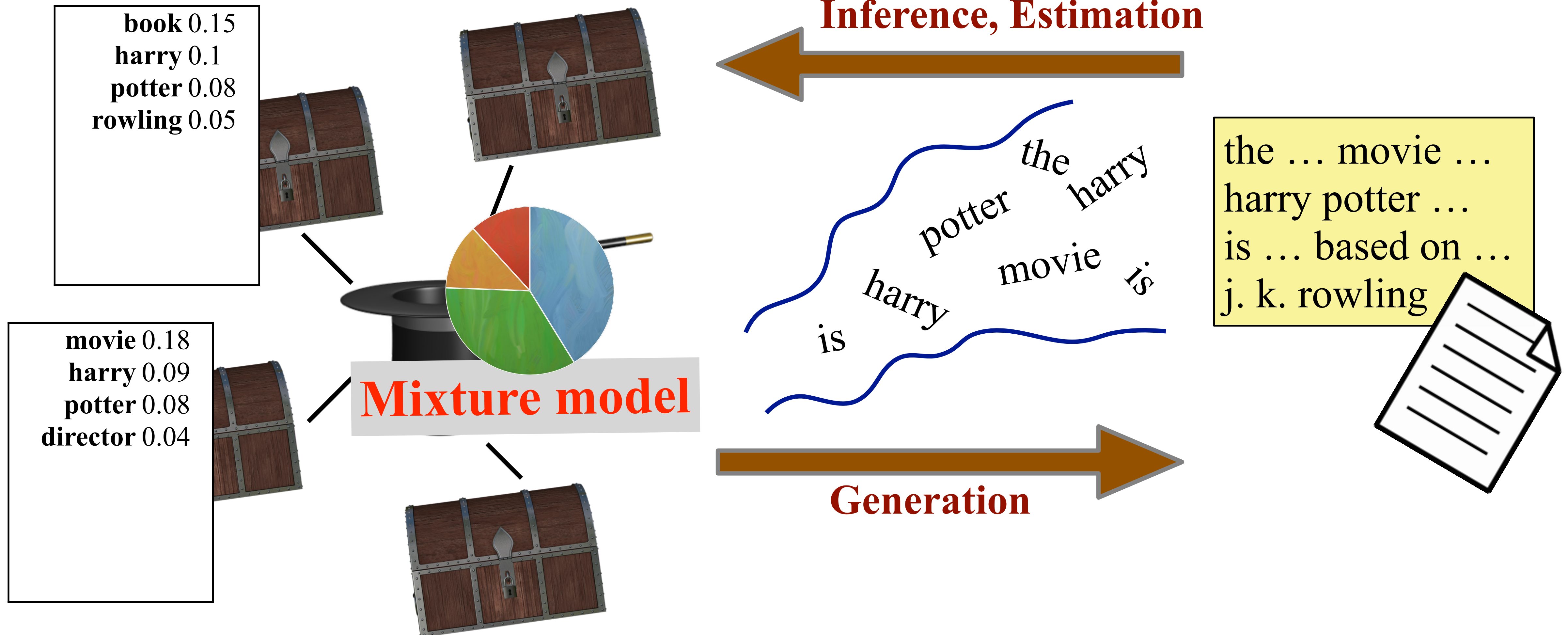
the ... movie ...
harry potter ...
is ... based on ...
j. k. rowling

Generation



Generative Models can be Complex

$\text{Pr}(\text{text} \mid \text{model})$



Latent Dirichlet Allocation (LDA)

- Generative model for a document d
 - Choose length of document d
 - Choose a mixture of topics for document d
 - Use a topic's multinomial distribution to output words to fill that topic's quota

Topic Modeling in Practice

- How many topics?
 - Finding or even guessing the number of topics is hard
- Interpreting topics
 - Topics are just word distributions
 - Making sense of words / generating labels is subjective

Topic Modeling: Summary

- Great tool for exploratory text analysis
 - What are the documents (tweets, reviews, news articles) about?
- Many tools available to do it effortlessly in Python

Working with LDA in Python

- Many packages available, such as gensim, lda
- Pre-processing text
 - Tokenize, normalize (lowercase)
 - Stop word removal
 - Stemming
- Convert tokenized documents to a document - term matrix
- Build LDA models on the doc-term matrix

Working with LDA in Python (2)

- **doc_set: set of pre-processed text documents**

```
import gensim
from gensim import corpora, models

dictionary = corpora.Dictionary(doc_set)
corpus = [dictionary.doc2bow(doc) for doc in doc_set]
ldamodel = gensim.models.ldamodel.LdaModel (corpus, num_topics=4,
id2word=dictionary, passes=50)
print(ldamodel.print_topics(num_topics=4, num_words=5))
```

- **ldamodel can also be used to find topic distribution of documents**

```
topic_dis = ldamodel[new_doc]
```


Take Home Concepts

- **Topic modeling is an exploratory tool frequently used for text mining**
- **Latent Dirichlet Allocation is a generative model used extensively for modeling large text corpora**
- **LDA can also be used as a feature selection technique for text classification and other tasks**